# Tutorial 4 –Experimental error (2)

## Background

- How to calculate the mean, SD and RSD for a set of data in the measurement?

$$\bar{x} = \frac{\Sigma_i(x_i)}{N} s = \qquad \text{SD} = \sqrt{\frac{\Sigma_i(x_i - \bar{x})^2}{N-1}} \qquad RSD = \frac{s}{\bar{x}} \times 100\%$$

- How to report the experimental result with different interests?

## Exercise

**T4-1.** A chemist performs analysis of chlorine in water, for each experiment, he performs 5 measurements. He repeats the experiment 3 times, as shown in the table:

| Exp. 1 | |
|---|---|
| **No. #** | **[$Cl^-$] ppm** |
| 1 | 1.032 |
| 2 | 0.998 |
| 3 | 1.014 |
| 4 | 1.026 |
| 5 | 1.029 |
| 6 | 1.003 |

| Exp. 2 | |
|---|---|
| **No. #** | **[$Cl^-$] ppm** |
| 1 | 0.990 |
| 2 | 0.970 |
| 3 | 0.993 |
| 4 | 0.967 |
| 5 | 0.968 |
| 6 | 1.014 |

| Exp. 3 | |
|---|---|
| **No. #** | **[$Cl^-$] ppm** |
| 1 | 1.010 |
| 2 | 0.958 |
| 3 | 1.008 |
| 4 | 0.998 |
| 5 | 1.032 |
| 6 | 0.950 |

a. Calculate the means ($\bar{x}$) for each experiment.

b. Calculate the standard deviation ($S$) for each experiment.

c. Which experiment had the best precision for the three experiments?

**a. Calculate the means:**

We don't know the real value of chlorine in water (μ, also called the population mean), therefore we estimate it by the means ($\bar{x} = \frac{\Sigma_i(x_i)}{N}$)

$\bar{x}_1 = \frac{1.032+0.998+1.014+1.026+1.029+1.003}{6}$ ppm = 1.017 ppm

$\bar{x}_2 = \frac{0.990+0.970+0.993+0.967+0.968+1.014}{6}$ ppm = 0.984 ppm

$\bar{x}_3 = \frac{1.010+0.958+1.008+0.998+1.032+0.950}{6}$ ppm = 0.993 ppm

b. **Calculate the standard deviation (SD or *s*):**

We also don't know the real standard deviation of the experiments σ (called the population standard deviation), therefore we estimate it by the standard deviation ($s = \sqrt{\frac{\Sigma_i(x_i-\bar{x})^2}{N-1}}$),

$s_1 = \sqrt{\frac{(1.032-1.017)^2+(0.998-1.017)^2+(1.014-1.017)^2+(1.026-1.017)^2+(1.029-1.017)^2+(1.003-1.017)^2}{6-1}}$ ppm

= 0.014 ppm ≈ 0.01 ppm

$s_2 = \sqrt{\frac{(0.990-0.984)^2+(0.970-0.984)^2+(0.993-0.984)^2+(0.967-0.984)^2+(0.968-0.984)^2+(1.014-0.984)^2}{6-1}}$ ppm

= 0.019 ppm ≈ 0.02 ppm

$s_3 = \sqrt{\frac{(1.010-0.993)^2+(0.958-0.993)^2+(1.008-0.993)^2+(0.998-0.993)^2+(1.032-0.993)^2+(0.950-0.993)^2}{6-1}}$ ppm

= 0.032 ppm ≈ 0.03 ppm

**Note:** we should round the SD to 1 significant figure and then present the result and the SD with the same decimal places and unit.

**c. Precision:**

Exp. 1 result: 1.02 (± 0.01) ppm

Exp. 2 result: 0.98 (± 0.02) ppm

Exp. 3 result: 0.99 (± 0.03) ppm

Precision refers to the reproducibility of a result, resulting from measuring many times of repetition with one method. The standard deviation (s) is the measure for precision. The lower value of s means good precision of the experiment. So, **Exp. 1** result of $s_1$ =0.01 ppm has the best precision.

**T4-2.** Find the relative standard deviation (RSD) for the numbers of 116.0, 97.9, 114.2, 106.8, and 108.3.

Mean = $\frac{116.0+\cdots+108.3}{5}$ = $108.6_4$

To avoid accumulating round-off errors, retain one more digit in the mean than was present in the original data.

$$\text{SD}=\sqrt{\frac{(116.0-108.64)^2+\cdots+(108.3-108.64)^2}{5-1}} = 7.1_4$$

The average and the standard deviation should both end at the same decimal place with the original data.

$$\text{RSD} = \frac{7.1_4}{108.6_4} \times 100\% = 6.5_7\% \approx 6.6\%$$

Finally, the RSD is 7% for consideration of S.F.

*Test yourself: how about the mean, SD and RSD for each of the five numbers 116.0, 97.9, 114.2, 106.8, 108.3 in the question is divided by 2?*

Answer: the mean and SD will be divided by 2, but the RSD is unchanged.

**T4-3.** The pressure in a system was measured many times. The following results were calculated:

$\bar{x}$ = 3.12356 $Mpa$

$S$D = 0.16789%

$n$ = 87

Rewrite the results according to the presentation rules you have learned (keeping a correct number of significant digits), for the following two cases:

a. You are interested in the measuring method.

b. You are interested in the results.

**a. Interested in the measuring method:**

As we learn in class, when focusing on the method or instrument, we need present the result as the following information:

- Standard deviation of measurements ($s$), with the unit of measurement
- Number of measurements ($N$)
- Mean in brackets ($\bar{x}$) or omission

Since the error given is a relative error ($\%e$), we need to convert it into absolute error ($e$) firstly.

$$\%e = \frac{e}{\bar{x}} \times 100\% \Longrightarrow e = \frac{\%e \times \bar{x}}{100\%} = \frac{0.16789\% \times 3.12356\ \text{MPa}}{100\%} = 0.005244\ \text{MPa}$$

**Note:** we must round the SD to 1 significant figure and then present the result and the SD with the same decimal places.

So, $e$ = 0.005 MPa

And the final results should be displayed as:

$s$ = 0.005 MPa

$N$ = 87

or $s$ = 0.005 MPa, $N$ = 87 ($\bar{x}$ = 3.124 $MPa$)

**b. Interested in the result:**

When focusing on the result, we need present the following information:

- Mean ($\bar{x}$)
- Standard deviation of the mean ($s_m$), with the unit of measurement
- Number of measurements ($N$)

We have get the absolute standard deviation ($s$), now we need to calculate the $s_m$ through the following formula:

$$s_m = \frac{s}{\sqrt{N}} = \frac{0.005244\ MPa}{\sqrt{87}} = 0.00056222\ \text{MPa} \Longrightarrow s_m = 0.0006\ \text{MPa}$$

The final results should be displayed as:

$\bar{x}$ = 3.1236 MPa

$s_m$ = 0.0006 MPa

$N$ = 87

or $\bar{x}$ = 3.1236 (± 0.0006) MPa, $N$ = 87

**T4-4.** Two people entered a star observatory; the first person said that god told him in a dream that there are 336 stars in $1m^2$ of sky. The second person did not believe it, and wanted to check whether the first person is indeed connected to god. Therefore, two of them counted the number of stars in 1 $m^2$ of sky 5 times together. These are their results:

| **Count number** | **Stars in 1 $m^2$ of sky** |
|---|---|
| 1 | 335 |
| 2 | 337 |
| 3 | 330 |
| 4 | 337 |
| 5 | 320 |

a. In the above experiment what is $\mu$ and what is $\bar{x}$?

b. Calculate the standard deviation for each person for these results. What does standard deviation represent?

The second person repeated the count for another two days and got the following results:

| **The second day** | |
|---|---|
| **Count number** | **Start in 1 $m^2$ of sky** |
| 1 | 324 |
| 2 | 341 |
| 3 | 331 |
| 4 | 334 |
| 5 | 331 |

| **The third day** | |
|---|---|
| **Count number** | **Start in 1 $m^2$ of sky** |
| 1 | 340 |
| 2 | 327 |
| 3 | 333 |
| 4 | 335 |
| 5 | 345 |

c. Calculate the mean number of stars in 1 $m^2$ of sky for every day, and calculate the mean value of stars in 1 $m^2$ of sky for the whole 3 days.

d. Calculate the standard deviation for the means. What does it represent?

e. Assuming the first person is right, in which day the precision is the highest and in which day the accuracy is the highest?

a. In the above experiment what is $\mu$ and what is $\bar{x}$?

$\mu = 336$ stars/m$^2$

$\bar{x} = 332$ stars/m$^2$

b. Calculate the standard deviation each person would have calculated for these results. What does each standard deviation represent?

Second person – 7.19 stars/m$^2$ ≈ 7 stars/m$^2$ $s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$

First person – 7.68 stars/m$^2$ ≈ 8 stars/m$^2$ $\sigma = \sqrt{\frac{\sum_i (x_i - \mu)^2}{n}}$

This standard deviation represents the random error in the experiment.

c. Calculate the mean number of stars in 1 $m^2$ of sky for every day, and calculate the mean value of stars in $1m^2$ of sky for the whole 3 days.

The first day – 332 stars/m$^2$

The second day – 332 stars/m$^2$

The third day – 336 stars/m$^2$

Whole 3 days – 333 stars/m$^2$

d. Calculate the standard deviation for the means. What does it represent?

Method 1 $s_m$ SD $(332\,,\ 332, 336) = 2.31$ stars/m$^2$) ≈ 2 stars/m$^2$

Method 2 $s_m \approx \frac{s\,(all\ data)}{\sqrt{n}} = \frac{s\,(15\ data)}{\sqrt{the\ number\ of\ repeat\ measurements}} = \frac{6.54}{\sqrt{15}}$

=1.69 stars/m$^2$ ≈ 2 stars/m$^2$

$s_m$ Represents the distribution of the averages of three measurements.

e. Assuming the first person is right, in which day the precision is the highest and in which day the accuracy is the highest?

$SD_1$ = 7.19 stars/m$^2$; $SD_2$ = 6.14 stars/m$^2$; $SD_3$ = 6.58 stars/m$^2$

Best precision – second day (lowest SD)

The first day – 332(±7) stars/m$^2$

The second day – 332(±6) stars/m$^2$

The third day – 336 (±7) stars/m$^2$

Best accuracy – Third day (High trueness to the $\mu$ with relative low SD)